

Fibre Science

Interactive concept, microscopic view, burning test, chemical solubility and examination handbook for Diploma Curriculum Revision 2026 — Course 1061.

Course at a glance

Course Code	1061
Contact Hours	60
Teaching Scheme	3:0:0:0
Credits	3
CIE	40
ESE	60
Bloom Level	Apply

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 1061 — Engineering Science**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

Official SITTTTR Revision 2026 syllabus reference. This page is a study handbook, not a substitute for the latest official notification or examination instruction.

C01

While origin tells us where a fiber comes from, **Chemical Classification** groups fibers by their chemical composition and structural polymers:

Chemical Class	Primary Chemical Component	Examples	Key Characteristics
Cellulosic	Cellulose polymer, $(C_6H_{10}O_5)_n$	Cotton, Jute, Linen, Flax, Hemp	High moisture regain, combustible, sensitive to mineral acids, resistant to dilute alkalis.
Protein	Polypeptides / Amino acids	Silk (Fibroin), Wool (Keratin)	Highly hygroscopic, self-extinguishing, sensitive to alkalis/bleach, moderate acid resistance.
Polyamide	Amide linkages ($-CO-NH-$)	Nylon 6, Nylon 6,6	High tenacity, excellent elastic recovery, thermoplastic.
Polyester	Ester linkages ($-CO-O-$)	Polyethylene Terephthalate (PET)	Highly crystalline, hydrophobic, wrinkle-resistant.
Polyacrylic	Acrylonitrile units	Acrylic, Modacrylic	Wool-like feel, lightweight, weather-resistant.
Mineral	Silicates, glass, carbon	Asbestos, Glass fibre	Non-combustible, extremely high thermal resistance.

Malayalam support: കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ ക്ലാസിഫിക്കേഷൻ Chemical Classification.

കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ, കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ (Cotton) കെമിക്കൽ (Jute) Cellulosic കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ കെമിക്കൽ Cellulose കെമിക്കൽ. കെമിക്കൽ (Silk) കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ (Wool) Protein കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ കെമിക്കൽ കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ (Polypeptides) കെമിക്കൽ ക്ലാസിഫിക്കേഷൻ.

1.3 Essential and Desirable Properties of Textile Fibres ↑

To be successfully converted into yarns and fabrics, a fibrous material must possess certain properties. These are classified into **Essential (Primary)** and **Desirable (Secondary)** properties:

1. Essential (Primary) Properties

- **Length to Width Ratio (Aspect Ratio):** Must be at least 1000:1 so that fibers can twist around each other and hold together.
- **Tenacity (Fibre Strength):** Sufficient breaking strength (at least 1.0 to 1.5 grams per denier) to withstand spinning tension.
- **Cohesiveness (Spinnability):** The ability of fibers to adhere/cohere to one another when twisted. Surface friction or crimp provides this.
- **Pliability (Flexibility):** Fibers must be soft and pliable enough to bend easily without breaking during twisting and knitting/weaving.
- **Uniformity:** Moderate consistency in physical dimensions (length and fineness) is required to produce uniform, defect-free yarns.

2. Desirable (Secondary) Properties

- **Moisture Regain / Absorption:** High absorption increases comfort, reduces static charge buildup, and facilitates dyeing.
- **Resiliency and Elasticity:** Ability to recover from deformation (wrinkling or stretching). Wool has high resiliency; cotton has low.
- **Luster:** The light-reflection property that gives a fiber its shine or matte look (e.g., Silk has highly prized triangular cross-sectional luster).
- **Chemical and Environmental Resistance:** Ability to withstand exposure to acids, alkalis, sunlight, heat, and microorganisms.
- **Flammability:** Behavior towards heat and flame (safety requirement).

Malayalam support: അത്യാവശ്യ ഗുണങ്ങൾ (Essential Properties) നന്നായ്കുന്ന ഗുണങ്ങൾ (Desirable Properties) എന്നിവയാണ്. അത്യാവശ്യ ഗുണങ്ങൾ: ശക്തി (strength), ഏകദൃഢത (flexibility), ചേർച്ച (cohesiveness) എന്നിവ.

1.4 Burning and Solubility Behavior of Textile Fibres ↑

Fibre identification relies heavily on physical response to flame and chemical dissolving behavior. This forms the scientific foundation for laboratory analysis.

1. Burning Test Behavior

Fibers behave distinctively when exposed to flame based on their chemical class:

Fibre Class	In Flame Response	Odor of Burning	Residue / Ash Character
Cellulosic (Cotton, Jute, Linen)	Burns quickly with yellow flame, does not melt. Continues burning after removal.	Burning paper / wood odor.	Light, fine, feather-like grey ash.
Protein (Silk, Wool)	Burns slowly with flickering flame, curls away. Self-extinguishes.	Burning hair / feathers odor.	Dark, brittle, crushable irregular bead or coal.
Synthetic (Polyester, Nylon)	Melts and shrinks away. Burns with black smoke.	Sweet chemical odor (Polyester) or celery-like odor (Nylon).	Hard, uncrushable, round dark/cream-colored plastic bead.

2. Selective Chemical Solubility

Chemical separation represents the most precise quantitative method to distinguish and isolate different fibers in blended yarns:

- **Cotton:** Dissolves in cold 70% Sulfuric Acid (H_2SO_4), but is highly resistant to hot dilute Sodium Hydroxide ($NaOH$).
- **Wool:** Dissolves completely in boiling 5% Sodium Hydroxide ($NaOH$), but is resistant to cold concentrated Sulfuric Acid.
- **Silk:** Dissolves in boiling 5% $NaOH$ and also in cold concentrated Hydrochloric Acid (HCl).
- **Polyester:** Outstanding resistance to acids and alkalis; dissolves in boiling Metacresol or boiling Nitrobenzene.

- **Nylon:** Dissolves in cold 85% Formic Acid (HCOOH).

[illegible]

1.5 Fibre Blends: Concept and Purpose

In modern textiles, fibers are rarely used in isolation. A **Fibre Blend** is a yarn made by mixing two or more different types of staple fibers or filaments together during spinning. A common example is *Polycot* (Polyester + Cotton) or *Polywool*.

Purpose of Blending Fibres

Fibre blending is done systematically to achieve characteristics that no single fiber can offer:

1 Improving Performance and Comfort:

Blending Polyester (strong, crease-resistant, quick-drying) with Cotton (absorbent, highly breathable, soft) creates a fabric that is comfortable to wear yet highly durable and easy to care for.

1. **Cost Optimization:** Mixing expensive premium fibers like Wool or Silk with inexpensive fibers like Polyester or Viscose makes the final garment highly affordable while retaining a luxury feel.
2. **Aesthetic Appeal and Texture:** Blending allows unique visual effects like cross-dyeing (different fibers absorbing different dyes) and creates custom textures or draping qualities.
3. **Processing Efficiency:** Blending highly static-prone synthetic fibers with natural conductive fibers reduces static electricity, making them easier to weave on automated high-speed looms.

Staple vs. Filament Fibres

- **Staple Fibres:** Short, limited-length fibers (measured in inches or millimeters). All natural fibers (except Silk) are staple fibers (e.g., Cotton is 1.5 to 5 cm long). They must be twisted together to form a yarn, giving it a fuzzy, textured appearance.
- **Filament Fibres:** Continuous, indefinite-length fibers (measured in meters or yards). Silk is the only natural filament fiber (ranging from 500 to 1500 meters). Synthetic fibers are extruded as continuous filaments which can be used directly or cut into staple lengths.

[illegible]

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Cotton Fibre - Structure, Properties & Identification

Delve into the botanical classification, microscopic cellular morphology, chemical reactions, and properties of cotton, the king of natural plant fibers.

11 Lecture Hours

10 Practical Hours

CO2

2.1 Classification and Botanical Varieties of Cotton ↑

Cotton is a natural seed-hair fiber. It grows in a protective capsule called a **boll** around the seeds of the cotton plant, which belongs to the genus **Gossypium** of the family **Malvaceae**.

Botanical Classification

There are four commercially cultivated species of cotton:

1 **Gossypium hirsutum:**

Known
as
Upland
Cotton.
Native
to
Central
America.
It
accounts
for
more
than
90%
of
global
cotton
production
due
to
its
high
yield
and
adaptability.

1. **Gossypium barbadense:** Known as Extra-Long Staple (ELS) cotton. Includes premium varieties like Egyptian, Pima, and Sea Island cottons. It yields extremely fine, long, and silky fibers of premium quality.
2. **Gossypium arboreum:** Known as Tree Cotton. Native to India and Pakistan. Highly resistant to pests and dry climates.
3. **Gossypium herbaceum:** Levant Cotton. Native to semi-arid regions of Africa and Asia. Short, coarse fibers.

Organic Cotton and BT Cotton

- **BT Cotton:** Genetically modified (GM) cotton containing genes from the bacterium *Bacillus thuringiensis*. The plant produces a toxin that selectively kills the devastating bollworm pest, reducing the need for chemical pesticides and dramatically increasing yield.
- **Organic Cotton:** Cotton grown without genetically modified seeds, synthetic pesticides, or chemical fertilizers. It promotes environmental biodiversity and soil health, fetching a premium market price.
- **Indian Hybrid Cottons:** India was the first country to develop commercial cotton hybrids (such as H4, MCU-5, and Varalaxmi) which combine high yield with long, fine staple lengths.

Major Producing Countries: China, India, United States, Brazil, and Pakistan.

Malayalam support: ഗോസ്സ്യം പ്ലാൻറേസിയം (Gossypium) പ്ലാൻറേസിയം (boll) പ്ലാൻറേസിയം പ്ലാൻറേസിയം. Gossypium പ്ലാൻറേസിയം പ്ലാൻറേസിയം. പ്ലാൻറേസിയം പ്ലാൻറേസിയം പ്ലാൻറേസിയം Gossypium hirsutum (Upland cotton) പ്ലാൻറേസിയം. പ്ലാൻറേസിയം പ്ലാൻറേസിയം പ്ലാൻറേസിയം പ്ലാൻറേസിയം BT Cotton-ൽ പ്ലാൻറേസിയം പ്ലാൻറേസിയം.

2.2 Structure and Chemical Composition of Cotton Fibre ↑

The microscopic structure of cotton is unique and easily identifiable under a light microscope. As the fiber dries, the central liquid-filled core collapses, causing the fiber to twist.

1. Microscopic Morphology

- **Longitudinal View:** Appears as a flat, twisted ribbon or collapsed hollow tube. These characteristic twists are called **convolutions**. Convolutions are crucial because they increase surface friction and cohesiveness, allowing flat fibers to be spun into strong yarns without slipping.
- **Cross-Sectional View:** Appears kidney-shaped, bean-shaped, or ear-like, with a central hollow canal called the **lumen**. The lumen is the dried-up remains of the living protoplasm cell that nourished the fiber during its growth inside the boll.

2. Chemical Composition of Raw Cotton

Raw cotton is highly pure cellulose, but it contains minor natural impurities that must be removed during scouring and bleaching:

- **Cellulose:** 90% to 94% (The main structural polymer, composed of repeating glucose units).
- **Water/Moisture:** 6% to 8% (Absorbed under atmospheric conditions).
- **Proteins, Waxy matter, and Pectins:** 1% to 2% (Forms the protective outer layer of the raw fiber, making raw cotton naturally water-repellent before scouring).
- **Mineral Ash:** 1%.

Malayalam support: കോമ്പ് ഓരോ കോമ്പത്തിനുള്ളിലുള്ള കോമ്പങ്ങൾ കോമ്പങ്ങൾ കോമ്പങ്ങൾ കോമ്പങ്ങൾ (convolutions) കോമ്പങ്ങൾ കോമ്പങ്ങൾ. കോമ്പങ്ങൾ കോമ്പങ്ങൾ കോമ്പങ്ങൾ 'Lumen' കോമ്പങ്ങൾ കോമ്പങ്ങൾ കോമ്പങ്ങൾ. കോമ്പങ്ങൾ കോമ്പങ്ങൾ കോമ്പങ്ങൾ 90 കോമ്പങ്ങൾ കോമ്പങ്ങൾ (Cellulose) കോമ്പങ്ങൾ കോമ്പങ്ങൾ. കോമ്പങ്ങൾ കോമ്പങ്ങൾ (wax), കോമ്പങ്ങൾ കോമ്പങ്ങൾ.

2.3 Physical and Chemical Properties of Cotton ↑

Cotton's properties dictate its ultimate end-use in apparel, home textiles, and industrial applications.

1. Physical Properties

- **Fibre Length:** Ranges from 1.5 cm to 6 cm. Longer staple lengths make smoother, stronger, and more valuable yarns.
- **Fibre Fineness:** Highly fine. Expressed in Micronaire value (weight in micrograms per inch).
- **Tenacity (Strength):** Moderate to high (3.0 to 5.0 g/denier). Unique property: **Cotton is 10% to 20% stronger when wet.** This is because water molecules enter the amorphous regions of the cellulose polymer, forming additional hydrogen bonds that stabilize the crystalline polymer chain. This makes cotton highly washable without damage.
- **Elongation and Resiliency:** Low elastic recovery (stretches only 3% to 7% before breaking) and low resiliency. This is why cotton garments wrinkle extremely easily.
- **Moisture Regain:** Standard moisture regain is 7.5% to 8.5%. Highly absorbent and comfortable on the skin.

2. Chemical Properties and Reactions

- **Effect of Acids:** Highly sensitive. Mineral acids (such as Hydrochloric acid HCl , Sulfuric acid H_2SO_4 , and Nitric acid HNO_3) hydrolyze the glycosidic linkages in the cellulose chain, causing rapid disintegration and dissolution of the fiber. Organic acids like citric or acetic acid are safe at low concentrations.
- **Effect of Alkalis:** Extremely resistant. Even boiling concentrated Sodium Hydroxide (NaOH) does not dissolve cotton in the absence of oxygen. When treated with cold strong NaOH (15%-20%) under tension, the fibers swell, lose their convolutions, become circular, and develop a highly lustrous appearance and increased dye absorption. This chemical process is called **Mercerization**.
- **Effect of Water and Bleach:** Resistant to water. Damaged by strong oxidizing agents (bleaches) if over-exposed, which converts cellulose to oxycellulose.
- **Effect of Sunlight:** Gradual yellowing and loss of strength due to photo-oxidation.
- **Effect of Microorganisms:** Susceptible to mildew, rot, and fungi in warm, damp, humid conditions.

[illegible]

Bast Fibres - Jute and Linen

Examine the extraction process, cellular composition, structure, and physical-chemical differences of bast fibers: Jute and Linen.

11 Lecture Hours

10 Practical Hours

CO3

3.1 Features and Extraction of Jute Fibres ↑

Jute is known as the "**Golden Fibre**" due to its golden-brown color and high commercial value. It is a bast fiber, meaning it is extracted from the phloem or inner bark of the stem of the **Corchorus** plant (species include *Corchorus capsularis* and *Corchorus olitorius*).

Jute Extraction Process (Retting and Stripping)

Extracting jute requires breaking down the pectins and gums holding the fiber bundles to the woody stem:

1

Harvesting:

The stalks are harvested when the flowers begin to shed.

- Retting:** The stems are tied into bundles and submerged in clean, slow-moving water for 10 to 20 days. Microbial action and anaerobic fermentation dissolve the sticky pectins and gums, releasing the fiber bundles from the woody core.
- Stripping (Extraction):** The retted stalks are taken out, and the fibers are manually stripped from the stem by beating the root ends and pulling the fiber strands.
- Washing and Drying:** The extracted fibers are washed in clean running water and hung on bamboo frames to dry in open sunlight for 2-3 days before baling.

Malayalam support: ജൂട്ട് (Golden Fibre) കർഷകർ ജൂട്ട് തണ്ടുകൾ മുറിക്കുക. തണ്ടുകൾക്ക് **Retting** എന്ന പ്രക്രിയയിൽ 10-20 ദിവസം വെള്ളത്തിൽ മുക്കേണ്ടതാണ്. ജൂട്ട് തണ്ടുകളിൽ നിന്ന് പെക്റ്റിൻ, ഗുമി എന്നിവ ലയിക്കുകയും കൈയൊഴുപ്പ് ഉപയോഗിച്ച് തണ്ടിൽ നിന്ന് കൈയൊഴുപ്പ് നീക്കം ചെയ്യുകയും ചെയ്യുന്നു. തണ്ടുകൾ വെള്ളത്തിൽ കഴുകി വെള്ളത്തിൽ വെക്കുക. തണ്ടുകൾ വെള്ളത്തിൽ വെക്കുക. തണ്ടുകൾ വെള്ളത്തിൽ വെക്കുക.

3.2 Structure and Chemical Composition of Jute ↑

Unlike cotton, jute is a highly lignified composite fiber, which explains its coarse texture, high stiffness, and low flexibility.

1. Microscopic Structure

- **Longitudinal View:** Appears as a bundle of smooth, overlapping polygonal cells with thick walls and irregular nodes. The fibers do not have convolutions.
- **Cross-Sectional View:** Polygonal or sharp-edged oval cells grouped in clusters, featuring a wide central hollow space (lumen) with varying diameter.

2. Chemical Composition of Jute

Jute contains a high percentage of hemicellulose and lignin, making it less pure than cotton:

- **α-Cellulose:** 58% to 63% (Provides primary tensile strength).
- **Hemicellulose:** 20% to 24% (Amorphous, highly sensitive to alkalis).
- **Lignin:** 12% to 15% (A complex natural polymer that acts as a cement, cementing the cellular fibrils together. It gives jute its stiffness, coarse feel, and brown color).
- **Fats, Wax, and Ash:** 1% to 2%.

Malayalam support: ജൂട്ടിന്റെ രാസസംയുക്തം പ്രധാനമായും 12-15% **Lignin**

ഉൾക്കൊള്ളുന്നു. ഈ ലിഗ്നിൻ ജൂട്ടിന്റെ കഠിനതയും ക്രൂരതയും നൽകുന്നു. ഇത് ജൂട്ടിന്റെ കോഴിപ്പരുത്തിയും, കറുപ്പും നൽകുന്നു. **polygonal** കോശങ്ങൾ കാണപ്പെടുന്നു.

3.3 Features and Extraction of Linen (Flax) Fibres ↑

Linen is one of the oldest textile fibers known. It is extracted from the bast of the **Flax** plant, botanically classified as **Linum usitatissimum**.

Linen Extraction Process

Linen extraction is highly specialized and involves several mechanical and biological steps:

1 Pulling:

Flax
plants
are
pulled
up
by
the
roots
rather

than
cut,
to
maximize
fiber
length.

1. **Retting:** The plants undergo biological retting (water retting, dew retting, or chemical retting) to break down the calcium pectate adhesive layer.
2. **Scutching:** The dried flax stems are passed through corrugated rollers to crush the brittle woody center. The stems are then beaten with a wooden paddle (scutching) to scrape away the woody fragments, releasing the long flax fibers.
3. **Hackling (Combing):** The fibers are pulled through series of metal combs (pins) to separate the short coarse fibers (called **tow**) from the long, fine, parallel fibers (called **line** fibers).

Malayalam support: ഫ്ലാക്സ് (Flax) നൂൽക്കുറിശ്ശി Linen നൂൽക്കുറിശ്ശി. നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി **Scutching** (നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി), **Hackling** (നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി) നൂൽക്കുറിശ്ശി. നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി 'Line' നൂൽക്കുറിശ്ശി നൂൽക്കുറിശ്ശി 'Tow' നൂൽക്കുറിശ്ശി.

3.4 Structure, Composition, and Comparison of Jute vs. Linen ↑

Linen has a much lower lignin content than jute, giving it a bright white-cream color, high flexibility, and luxurious feel.

1. Microscopic Structure of Linen

- **Longitudinal View:** Appears as straight, cylindrical glass-like rods with characteristic swellings called **nodes** or "cross-markings" that resemble bamboo. Convolutions are absent.
- **Cross-Sectional View:** Well-defined polygons (typically 5 to 6 sided) clustered in neat bundles. The lumen is very narrow and central.

2. Chemical Composition of Linen

- **Cellulose:** 70% to 75% (Provides high strength).
- **Hemicellulose and Pectins:** 15% to 20%.
- **Lignin:** 2% to 5% (Much lower than Jute; gives strength without severe stiffness).

3. Chemical Reactions of Bast Fibres (Jute & Linen)

- **Effect of Acids:** Susceptible to hydrolysis by strong mineral acids. Both disintegrate rapidly.
- **Effect of Alkalis:** Strong alkalis dissolve the hemicellulose binding materials in Jute, causing the fiber bundles to separate and lose strength (called *splitting*). Linen is highly resistant to dilute alkalis but suffers damage under extreme conditions.

Malayalam support: മലയാളം സാധനങ്ങൾ ഉൾക്കൊള്ളുന്ന പേജുകൾക്ക് മലയാളം (bamboo) സാധനങ്ങൾ ഉൾക്കൊള്ളുന്ന (nodes) പേജുകൾ. മലയാളം സാധനങ്ങൾ ഉൾക്കൊള്ളുന്ന പേജുകൾക്ക് (2-5%) മലയാളം സാധനങ്ങൾ ഉൾക്കൊള്ളുന്ന പേജുകൾ ഉൾക്കൊള്ളുന്ന പേജുകൾ. മലയാളം (Jute) സാധനങ്ങൾ 15% മലയാളം സാധനങ്ങൾ ഉൾക്കൊള്ളുന്ന പേജുകൾ ഉൾക്കൊള്ളുന്ന പേജുകൾ.

Natural Protein Fibres - Silk and Wool

Study the biology, sericulture, chemical composition, properties, processing, and microscopic features of animal protein fibers: Silk and Wool.

12 Lecture Hours

9 Practical Hours

CO4

4.1 Silk: Biology, Sericulture, and Lifecycle of Silkworm ↑

Known as the "**Queen of Textiles**", Silk is a natural animal protein filament. It is secreted by the larvae of the silk moth during the construction of their protective cocoon.

1. Sericulture and Lifecycle

Sericulture is the commercial rearing of silkworms for the production of silk. The most widely used species is the domesticated Mulberry silkworm, **Bombyx mori**.

The life cycle of the silkworm consists of four distinct biological stages:

1 Egg (Graine):

The adult female moth lays hundreds of tiny eggs on mulberry leaves.

1. **Larva / Caterpillar:** The eggs hatch into larvae which feed voraciously on fresh mulberry leaves for 25 to 35 days, growing through 5 instars (molting phases). During this phase, the larva develops two large silk glands near its head.
2. **Pupa / Cocoon Formation:** When fully grown, the caterpillar spins a protective cocoon around itself by secreting a continuous double filament from its spinneret. This filament is composed of fibroin proteins held together by sericin gum. The larva transforms into a pupa inside this cocoon.
3. **Adult Moth:** If allowed to live, the pupa emerges as an adult moth in 10-14 days by secreting an alkaline fluid to dissolve the cocoon, breaking the continuous silk filament.

2. Varieties of Silk

- **Mulberry Silk:** Produced by *Bombyx mori*. Accounts for 90% of global production. Highly fine, uniform, and white-cream.
- **Wild/Non-Mulberry Silks:**
 - *Tussar Silk:* Copperish-bronze color, extracted from oak or terminalia-feeding worms.
 - *Muga Silk:* Rich golden luster, extremely durable, produced exclusively in Assam, India.
 - *Eri Silk:* Creamy white, spun-silk (since the cocoon is open-ended and the moth emerges naturally).

[illegible]

4.2 Silk Processing: Preparation, Degumming, and Weighting

Raw cocoons must undergo processing to extract the clean, continuous filament.

1. Reeling and Preparation of Raw Silk

1 Stifling:

The live pupae inside the cocoons are killed using dry heat or steam. This prevents the moth from emerging and breaking the filament.

1. **Boiling / Sorting:** Cocoons are sorted and boiled in hot water to soften the sticky **sericin** (silk gum).

2. **Brushing and Reeling:** The loose outer fibers are brushed to find the single continuous end of the filament. 4 to 8 filaments are combined, passed through a guide, and wound onto a reel to form a single strong thread of raw silk.

2. Chemical Composition of Raw Silk

- **Fibroin:** 70% to 75% (The insoluble structural protein core, crystalline in nature).
- **Sericin:** 20% to 25% (The gummy, amorphous protein sheath that glues the two fibroin filaments together).
- **Waxes and Inorganic Matter:** 1% to 2%.

3. Degumming and Weighting

- **Degumming:** The raw silk yarn is boiled in mild soap solution to dissolve and strip away the yellow sericin gum. This reveals the highly lustrous, soft, drapeable fibroin underneath. Silk loses about 25% of its weight during this process.
- **Weighting:** To compensate for the 25% weight loss during degumming, silk is treated with metallic salts (such as stannic chloride) which are absorbed by the fiber. Excessive weighting makes silk brittle and prone to splitting under sunlight.

4. Microscopic Appearance

- **Raw Silk:** Appears as a double filament (two fibroins) bound together by an irregular coating of sericin.
- **Degummed Silk:** Appears as smooth, regular, transparent triangular prisms that reflect light at unique angles, giving silk its characteristic luster.

[illegible]

4.3 Wool: Production, Varieties, and Shearing

Wool is a natural protein fiber obtained from the fleece of sheep, goats, or camelids. It belongs to the **hair-follicle** class of fibers.

1. Shearing and Varieties

- **Shearing:** The process of shaving off the fleece of the sheep in one continuous piece using mechanical clippers. This is usually done once a year in spring.
- **Varieties of Wool:**
 - *Merino Wool:* Extracted from Merino sheep. The finest, softest, and most highly prized wool with dense crimps.
 - *Coarse / Carpet Wool:* Coarser fibers used for rugs and carpets.
 - *Lamb's Wool:* Extremely soft wool shorn from sheep less than eight months old.

2. Basic Terminology

- **Crimp:** The natural wavy or spring-like waviness along the wool fiber. Crimp is crucial because it traps air pockets, providing exceptional thermal insulation and natural elasticity.
- **Felting:** The unique property of wool fibers to interlock and entangle permanently under the influence of friction, moisture, heat, and soap. This occurs due to the microscopic overlapping scales on the fiber surface.

Malayalam support: Shearing എന്നാണ് മൃഗങ്ങളെ കീറുന്ന പ്രക്രിയയെ വിളിക്കുന്നത്. Crimp എന്നതാണ് കീറുന്നതിനുള്ള കാര്യം. ഈ കീറുന്നതിനുള്ള കാര്യം (thermal insulation) നൽകുന്നു. Felting എന്നതാണ് കീറുന്നതിനുള്ള കാര്യം.

4.4 Wool Structure and Processing: Scouring, Carbonization, and Chlorination ↑

Raw wool shorn from sheep is highly contaminated with grease, sweat, dirt, and burrs. It must undergo extensive chemical processing.

1. Wool Processing Steps

1 Scouring:

Raw **Lanolin**
wool
is
washed
in
hot
alkaline
water
containing
soap
or
synthetic
detergents.
This
emulsifies
and
removes

(wool
grease/wax)
and
dried
sweat
(suint).
Lanolin
is
recovered
as a

highly
valuable
byproduct
for
cosmetics.

1. **Carbonization:** To remove cellulosic vegetable impurities (burrs, seeds, leaves) that cannot be scoured out, wool is treated with dilute sulfuric acid (H_2SO_4) followed by baking at $100^\circ C$. The acid carbonizes (burns) the cellulose into brittle charcoal, which is crushed and blown away, leaving the acid-resistant wool undamaged.
2. **Chlorination:** Treatment of wool with chlorine gas or hypochlorite solution to etch or coat the microscopic scales. This reduces wool's tendency to shrink and felt, making the wool washable in household washing machines.

2. Microscopic Structure of Wool

- **Longitudinal View:** Appears as a cylindrical tube completely covered with overlapping microscopic scales called **cuticle scales**, pointing towards the fiber tip.
- **Cross-Sectional View:** Round or slightly oval, composed of three distinct cellular layers:
 - *Cuticle:* The outer protective layer of scales.
 - *Cortex:* The main structural body containing ortho-cortex and para-cortex cells (which swell differently, producing the natural crimp).
 - *Medulla:* A hollow canal of dead cells running down the center of coarse fibers (absent in fine Merino wool).

3. Chemical Composition of Wool

Wool is composed of the protein **Keratin**. It is highly unique because it contains 3% to 4% sulfur, primarily in the form of **cystine disulfide linkages** ($-S-S-$ crosslinks) which act as internal springs, providing wool with outstanding elasticity and recovery.

Malayalam support: മലയാളത്തിൽ ഈ വിഷയത്തെക്കുറിച്ച് കൂടുതൽ അറിയാൻ [ഇവിടെ](#) ക്ലിക്ക് ചെയ്യുക.

മലയാളത്തിൽ ഈ വിഷയത്തെക്കുറിച്ച് കൂടുതൽ അറിയാൻ [ഇവിടെ](#) ക്ലിക്ക് ചെയ്യുക. (Lanolin) മലയാളത്തിൽ ഈ വിഷയത്തെക്കുറിച്ച് കൂടുതൽ അറിയാൻ [ഇവിടെ](#) ക്ലിക്ക് ചെയ്യുക.

Scouring. മലയാളത്തിൽ ഈ വിഷയത്തെക്കുറിച്ച് കൂടുതൽ അറിയാൻ [ഇവിടെ](#) ക്ലിക്ക് ചെയ്യുക. **Carbonization.** മലയാളത്തിൽ ഈ വിഷയത്തെക്കുറിച്ച് കൂടുതൽ അറിയാൻ [ഇവിടെ](#) ക്ലിക്ക് ചെയ്യുക.

Chlorination മലയാളത്തിൽ ഈ വിഷയത്തെക്കുറിച്ച് കൂടുതൽ അറിയാൻ [ഇവിടെ](#) ക്ലിക്ക് ചെയ്യുക.

4.5 Physical and Chemical Properties of Silk & Wool ↑

Animal fibers exhibit completely different characteristics compared to plant-based cellulosic fibers.

1. Comparison of Physical Properties

- **Fibre Form:** Silk is a continuous filament (up to 1500m); Wool is a short crimpy staple fiber (5 to 15 cm).
- **Tenacity:** Silk is extremely strong (4.0 to 5.0 g/d, comparable to steel wire of same diameter). Wool has very low tenacity (1.0 to 1.5 g/d, and drops by 20% when wet due to hydrogen bond breaking).

Figure 2: Jute microscopic views. Highly lignified polygonal cross-sectional cells clustered in neat bundles.

ngitudinal View: Cross-Nodes (Bamboo Structu

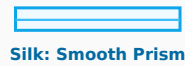


5-6 Sided Polygons (Tiny Lumen)

Figure 3: Linen (Flax) micro-morphology showing regular node marks (joints) and polygonal cross-section with highly constricted lumen.



Wool: Scale sheath



Silk: Smooth Prism



Round (Cortex/Medulla)



Triangular Cross-Section

Figure 4: Microscopic structure comparison. Wool fibers possess outer protective scales. Silk filaments display smooth triangular prism cross sections giving high luster.

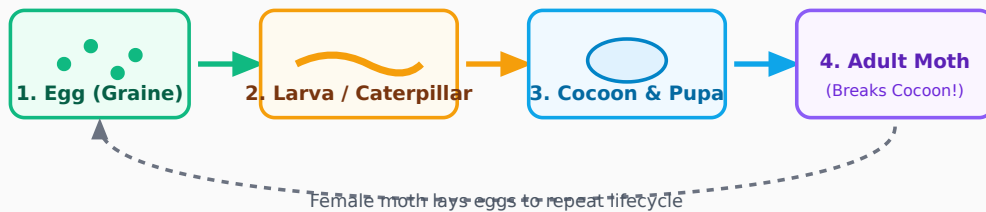


Figure 5: Biological lifecycle of Bombyx mori silkworm. Commercial sericulture harvests/stifles the cocoon at Stage 3 to preserve the continuous filament.

VIRTUAL TEXTILE LABORATORY

Interactive Fibre Identification Simulators

Hone your analytical skills by testing virtual textile fiber samples under different physical and chemical constraints.

1. Flame & Burning Test Simulator

Select a fiber sample below to simulate its burning action near a flame, inside a flame, and analyze its distinctive odor and ash residue.

Select Fibre Sample:

Simulation Status: Awaiting sample

2. Chemical Dissolution & Blend Calculator

Simulate a selective chemical dissolution test for binary fiber blends (e.g., Polyester-Cotton) to determine the percentage composition of each constituent.

Initial Dry Weight (g): Select Solvent Reagent:

Blend Combination:

selection...

Calculated Composition: Enter parameters and run simulation.

3. Virtual Light Microscope Viewer (10x and 40x zoom)

Toggle magnification and select any natural fiber to visualize its microscopic longitudinal structure and cross-sectional morphology under the lens.

Select Fibre to Inspect:

Objective Lens Zoom:

10x Low Power 40x High Power



Select a fiber and lens magnification, then click **Focus Microscope** to generate simulated high-resolution optical morphology.

PRACTICUM MANUAL

Fibre Science Practical Lab Exercises

Step-by-step procedures, observation recording structures, chemical proportions, and setup directions for natural fiber laboratory exploration.

Experiment 1: Flame Identification of Fibres

Aim: To distinguish textile fiber samples (Cotton, Silk, Wool, Jute, Nylon, Polyester) using burning tests.

Apparatus: Bunsen burner, crucible tongs, porcelain dish, matches, sample fibers.

Procedure:

1

Hold a small cluster of the fiber sample with tongs.

1. Slowly approach the flame from the side to observe behavior *near flame*.
2. Place the sample directly into the flame to observe behavior *in flame*.
3. Remove the sample from the flame to observe if it self-extinguishes.
4. Note the burning odor and inspect the cooled ash residue left behind.

Lab Note: Cellulose 纖維素是由葡萄糖單位組成的長鏈聚合物。Protein 蛋白質是由氨基酸組成的長鏈聚合物。Synthetic 合成纖維是由人工合成的聚合物。Natural 天然纖維是由天然材料製成的纖維。

Experiment 2: Microscopic Examination (10x / 40x)

Aim: To prepare temporary dry mount slides and analyze the longitudinal surface features of Cotton, Jute, Linen, Silk, and Wool under a light microscope.

Apparatus: Light microscope, glass slides, cover slips, dissecting needles, glycerine, fibers.

Procedure:

1. Separate a single, ultra-fine fiber strand using needles.

- 1. Place the fiber longitudinally on a clean glass slide.
- 2. Add one drop of water or glycerine to minimize refraction under high power.
- 3. Gently lower the cover slip to prevent air bubbles.
- 4. Examine under 10x magnification for fiber alignment, then switch to 40x for scale and node structures.

Lab Note: Convolutions-
Polygonal bamboo-cuticle scales (cuticle scales)

Experiment 3: Selective Solubility Tests

Aim: To differentiate textile fibers chemically using selective solubility reagents.

Reagents: 70% Sulfuric Acid (H_2SO_4), boiling 5% Sodium Hydroxide ($NaOH$), 85% Formic Acid, Acetone.

Procedure: Place small fiber snippets into test tubes. Add the reagent, agitate, and observe if dissolution happens in cold or hot conditions. Refer to the matrix below:

Fibre	Cold H_2SO_4 (70%)	Boiling $NaOH$ (5%)	Cold Formic Acid (85%)
Cotton	Soluble	Insoluble	Insoluble
Wool	Insoluble	Soluble	Insoluble
Nylon	Soluble	Insoluble	Soluble

Experiment 4: Binary Blend Quantitative Separation

Aim: To separate and calculate the dry weight percentages of a Polyester-Cotton (Polycot) blended sample.

Procedure: Weigh a clean, dry blended specimen (W_1). Immerse in cold 70% H_2SO_4 with agitation for 30 minutes to dissolve the cotton component. Filter the remaining polyester fibers using a sintered glass crucible. Wash, dry completely in an oven, and weigh the residue (W_2).

Cotton $\% = \frac{W_1 - W_2}{W_1} \times 100$, $\quad$ Polyester $\% = \frac{W_2}{W_1} \times 100$

Lab Note: ശുദ്ധീകരിച്ച കോട്ടൺ ഫൈബർ സാമ്പിൾ ഉപയോഗിച്ച് കോട്ടൺ ഫൈബർ ഓക്സൈഡേഷൻ പരീക്ഷണം നടത്തുക. ഓക്സൈഡേഷൻ പരീക്ഷണം നടത്തുന്നതിന് മുമ്പ് ഡിസിക്കേറ്റർ ഉപയോഗിക്കുക.

FIBRE QUESTION BANK

Module-wise Expected Questions with Answers

A comprehensive compilation of highly structured short, medium, and descriptive academic questions with precise model answers.

Q1. Define a textile fibre. What is the minimum aspect ratio required?

Answer: A textile fiber is a unit of matter characterized by flexibility, fineness, and high ratio of length-to-thickness (at least 1000:1) which renders it spin-spinnable into yarn. It must possess adequate tensile strength and cohesiveness.

Malayalam: കോട്ടൺ ഫൈബർ 1000 മടങ്ങിലധികം നീളം വീതിയുടെ അനുപാതം ഉള്ളതാണ്, ഇത് സ്പിൻ-സ്പിൻ ചെയ്യാൻ സാധിക്കും. കോട്ടൺ ഫൈബർക്ക് കൂടുതൽ ടെൻസൈൽ സ്ട്രെങ്ത് ഉണ്ടായിരിക്കണം. Textile fibre ന്റെ ഏതെങ്കിലും ഏതെങ്കിലും.

Q2. Explain the purpose and importance of blending fibers in the industry.

Answer: Blending is done to:

1. Combine properties of synthetic and natural fibers (e.g. blend strong crease-free Polyester with comfortable absorbent Cotton).
2. Reduce manufacturing costs of fabrics.
3. Introduce multi-color and visual textures during piece dyeing.
4. Minimize static electricity build-up during weaving.

Q3. Why is cotton 10% to 20% stronger when wet than dry?

Answer: When cotton is wet, water molecules penetrate the amorphous regions of cellulose, forming new, stable hydrogen bonds with hydroxyl groups. This relieves internal stresses, aligns the crystalline polymer chains, and spreads tensile load uniformly, increasing tenacity.

Malayalam: കോട്ടൺ ഫൈബർ ഉറപ്പായതിനുള്ള കാരണം Cellulose ന്റെ അമോർഫസ് പ്രദേശങ്ങളിൽ വെള്ള തന്മാത്രകൾ പെനീട്രേറ്റ് ചെയ്ത് ഹൈഡ്രജൻ ബന്ധം രൂപപ്പെടുത്തുന്നു. ഇത് ആന്തരിക സ്ട്രെസ്സുകൾ കുറയ്ക്കുന്നു, ക്രിസ്റ്റലൈൻ പോളിമർ ചെയിനുകൾ അലൈൻ ചെയ്യുന്നു, ടെൻസൈൽ ലോഡ് യൂണിഫോം ആയി വിതരണം ചെയ്യുന്നു, ഇത് ടെനാസിറ്റി വർദ്ധിപ്പിക്കുന്നു.

Q4. Explain what Mercerization is and its benefits.

Answer: Mercerization is the treatment of cotton yarns or fabrics with cold 15-20% Sodium Hydroxide (NaOH) under tension. This causes the fibers to swell, losing convolutions and turning round, which increases luster, dye absorption, strength, and dimensional stability.

Q5. Detail the biological process of jute Retting.

Answer: Retting is a biological fermentation process. Jute stems are bundled and kept underwater for 10-20 days. Anaerobic bacteria and microorganisms secrete enzymes that consume pectins, gums, and hemicellulose holding the bark fiber bundles to the inner woody stick, releasing pure jute strands.

Q6. How can you differentiate Jute and Linen fibers under a microscope?

Answer: Under a microscope, Jute fibers appear straight with thick walls and wide variable lumen without nodes. Linen (Flax) fibers appear straight and cylindrical with characteristic nodes (bamboo-like joints) crossing the fiber surface. Linen has a much narrower lumen than Jute.

Q7. Define shearing and felting of wool.

Answer:

Shearing: The mechanical process of shaving off the entire wool fleece from live sheep using professional clippers.

Felting: The biological interlocking of wool cuticle scales when subjected to moisture, heat, friction, and alkaline soap, creating a dense, non-woven mat of fibers.

Q8. Why do boiling alkalis dissolve animal fibers but leave plant fibers intact?

Answer: Silk (fibroin) and Wool (keratin) are made of protein polypeptide chains. Peptide linkages and disulfide bridges are highly vulnerable to hydrolysis by hydroxyl ions (OH^-), breaking the protein backbone into soluble amino acids. Cellulosic plant fibers (Cotton, Linen) are highly resistant to basic hydrolysis.

Fibre Science Course 1061 Theory Practice Paper

Time: 2.5 Hours | Max Marks: 60 (Strict SITTTTR Revision 2026 Curriculum Blueprint.
Questions mapped from Module I to IV with internal choices.)

PART A (Answer all questions. Each question carries 1 mark)

1. Define the term 'convolutions' in cotton. [1 Mark]
2. What is the primary chemical constituent of raw silk fiber? [1 Mark]
3. Identify one natural mineral fiber. [1 Mark]
4. State the botanical name of the Flax plant. [1 Mark]
5. What is the standard moisture regain percentage of dry wool? [1 Mark]
6. Define the term 'Lumen'. [1 Mark]
7. What byproduct is extracted during wool scouring? [1 Mark]
8. Identify the chemical compound used to dissolve Nylon fibers selectively in the laboratory. [1 Mark]

PART B (Answer any 8 of 12 questions. Each question carries 3 marks)

9. Distinguish between staple fibers and filament fibers with examples. [3 Marks]
10. List three essential (primary) properties of textile fibers and explain why they are critical. [3 Marks]
11. Write a short note on BT Cotton, explaining its genetic insect-resistance. [3 Marks]
12. Describe the chemical composition of raw cotton fiber, outlining the cellulosic content. [3 Marks]
13. Outline the manual process of 'stripping' jute fibers from retted stems. [3 Marks]
14. Why are mineral acids highly damaging to cotton fibers? Explain the reaction. [3 Marks]
15. Differentiate between the reeled 'line' flax fibers and coarse 'tow' flax fibers. [3 Marks]
16. Explain the method of degumming silk, detailing what protein is dissolved. [3 Marks]
17. What is 'Lanolin'? Describe its commercial recovery during raw wool processing. [3 Marks]
18. Explain how a burning test can differentiate a wool sample from an acrylic fiber. [3 Marks]

PART C (Answer one full question from each module. Each question carries 7 marks)

19.A. Classify textile fibers extensively based on origin. Draw a complete block flow schematic tree. **[7 Marks]**

OR

19.B. Discuss the chemical solubility of cotton, wool, nylon, and polyester in H₂SO₄, NaOH, and Formic Acid reagents. Summarize inside a neat tabular matrix. **[7 Marks]**

MODULE II

20.A. Draw a detailed, neat cross-sectional and longitudinal sketch of a cotton fiber. Explain convolutions and lumen in detail. **[7 Marks]**

OR

20.B. Contrast the physical and chemical properties of cotton. Explain how mercerization increases the luster and dye uptake. **[7 Marks]**

MODULE III

21.A. Explain step-by-step how flax fibers are extracted from the stem of *Linum usitatissimum*, describing retting, scutching, and hackling. **[7 Marks]**

OR

21.B. Compare Jute and Linen fibers thoroughly with respect to chemical composition, physical attributes, microscopic appearance, and end-uses. **[7 Marks]**

MODULE IV

22.A. Outline the biological life cycle of *Bombyx mori* silkworm. Explain the commercial production and harvesting of silk from raw cocoons. **[7 Marks]**

OR

22.B. Detail the scouring, carbonizing, and chlorination of wool. Explain the chemical reasons why wool displays high resiliency and crimp. **[7 Marks]**

MODEL SOLUTIONS

Master Answer Key and Marking Guide

1. Convolutions: Natural twists along the longitudinal ribbon structure of cotton. They increase fiber friction and cohesiveness, enabling spinning.

2. Fibroin: The inner crystalline core protein. (Sericin is the outer sticky gum protein).

3. Asbestos: Natural silicate mineral fibers, highly resistant to fire.

4. Botanical Name: *Linum usitatissimum* (family Linaceae).

5. Moisture Regain of Wool: Standard value is 15% to 18% (highly hygroscopic).

6. Lumen: Central hollow canal running down cellulosic plant fibers, previously acting as the living protoplasm core during cell growth.

7. Lanolin: Raw wool grease/wax, widely recovered and purified for cosmetic and pharmaceutical ointments.

8. Nylon solvent: Cold 85% Formic Acid (HCOOH) dissolves polyamides

selectively.

9. Staple vs Filament: Staple fibers are short, limited-length (e.g. Cotton, Wool); Filaments are long, continuous strands of indefinite length (e.g. Silk, Nylon).

10. Essential properties: 1. Aspect ratio > 1000:1 (spinnability), 2. Tenacity > 1 g/d (weaving strength), 3. Flexibility (pliability to bend into knitted/woven loops).

19.A/B scoring: Award marks for detailed taxonomic flowcharts (Vegetable, Animal, Mineral) and exact acid/alkali concentration matrices.

22.A/B scoring: Detail both sericin-to-fibroin ratios (25/75) for silk, and cystine disulfide sulfur bonds (Keratin) inside the cortex layers for wool.

INDEPENDENT STUDY

Suggested Student Activities for Self-Learning

Suggested active-learning curriculum assignments, industrial research tasks, and reporting formats to fulfill the 90-hour self-learning credits.

Module	Weekly Activity Description	Expected Outcome / Deliverable	Hours
Module I	Create a detailed flowchart with written explanations tracing the journey of one natural fiber from raw seed/source to a finished commercial garment.	Visual value chain report.	6 Hours
Module I	Study the agricultural characteristics, cultivation, and technical properties of Sisal leaf fibers.	Brief note on leaf fiber applications.	6 Hours
Module I	Write a critical essay comparing the ecological footprint and environmental sustainability of natural fibers vs. synthetic microplastics.	Sustainability research report.	6 Hours
Module II	Examine the technical characteristics of organic cotton, non-traditional medical cotton products, and non-woven hygiene wipes.	Comparative application notes.	6 Hours
Module III	Investigate the role and economic contribution of Jute crop export markets to the national and agricultural economy of India.	Economic analysis report.	6 Hours
Module IV	Explore the concept of ethical silk harvesting (Ahimsa Silk/Eri Silk) and organic wool rearing guidelines.	Ethical practices research paper.	6 Hours

TERMINOLOGY REFERENCE

Textile Fiber Glossary

- Aspect Ratio: The ratio of fiber length to diameter. Must exceed 1000:1 to spin yarn successfully.
- Bast Fibre: Cellulose fibers extracted from the phloem/bark of dicotyledonous plant stems (e.g. Jute, Flax).
- Convolutions: Natural spiral twists on the longitudinal ribbon surface of collapsed cotton fibers.
- Keratin: The sulfur-rich structural protein containing cystine crosslinks that forms wool

and animal hair.

Fibroin: The insoluble structural protein core making up 75% of raw silk filament.

Sericin: The hot-water-soluble protein gum that sheaths and cements silk fibroin filaments.

Mercerization: Alkali modification of cotton with cold NaOH under tension to boost luster, absorption, and strength.

Retting: Microbial fermentation under water to digest gums/pectins from flax or jute stems, freeing fibers.

Lanolin: Valuable natural grease recovered from raw wool wash liquor during commercial scouring.

OFFICIAL RESOURCES

Syllabus Textbooks & Reference Assets

Official SITTTR Prescribed Textbooks

1. **Text Book of Fibre Science and Technology** — S. P. Mishra (New Age International).
2. **Textile Science** — Gohl & Valensky (Mahajan Publishers, Ahmedabad).

Reference Literature

- *Textile Science* — J. T. Marsh (Read Books).
- *Fundamentals of Fibre Science* — L. C. Sonntag, P. M. Joseph (John Wiley & Sons).
- *Handbook of Textile Fibres* — J. Gordon Cook (Woodhead Publishing).

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